

Algebraic invariants associated to graph coloring

Raquel Melgar (joint work with Jean-Christophe Aval)

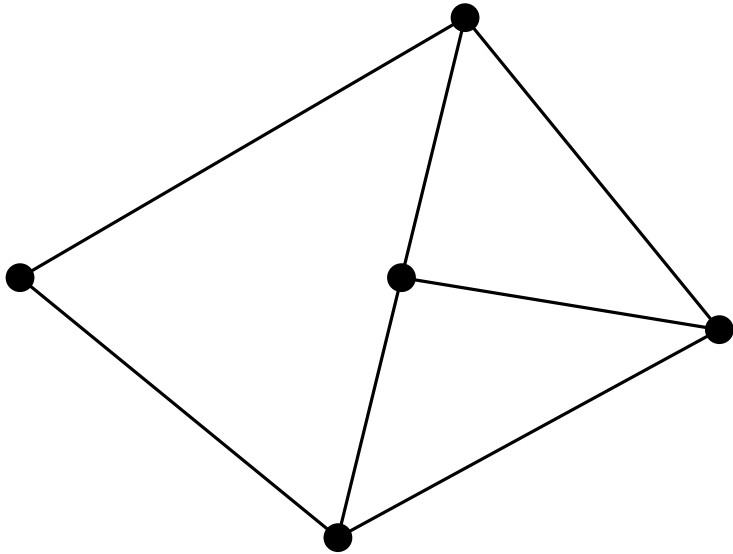
LaBRI (Université de Bordeaux)

16th January 2025, RSME's VIIth congress of Young
Researchers, Bilbao

First definitions: Graph coloring

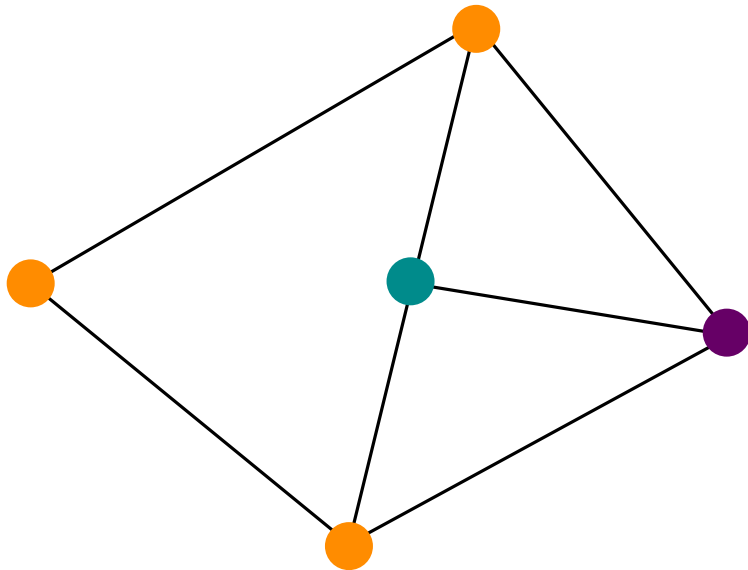
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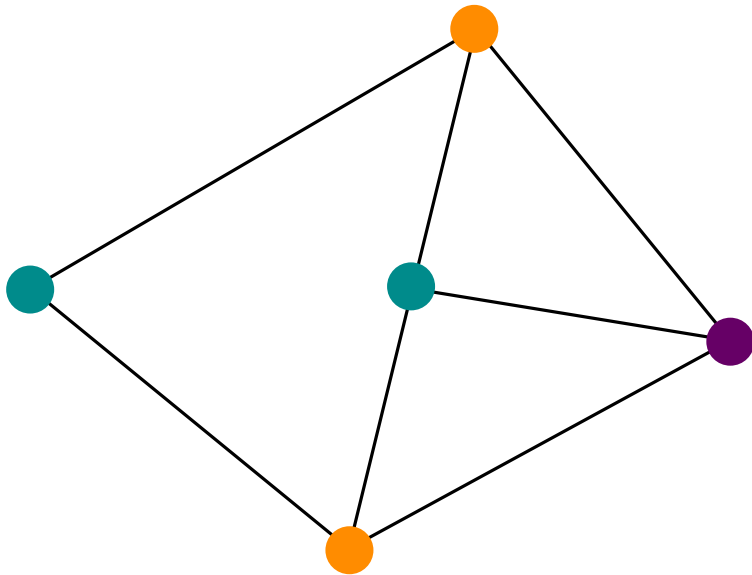
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This is **not** a proper coloring of Γ

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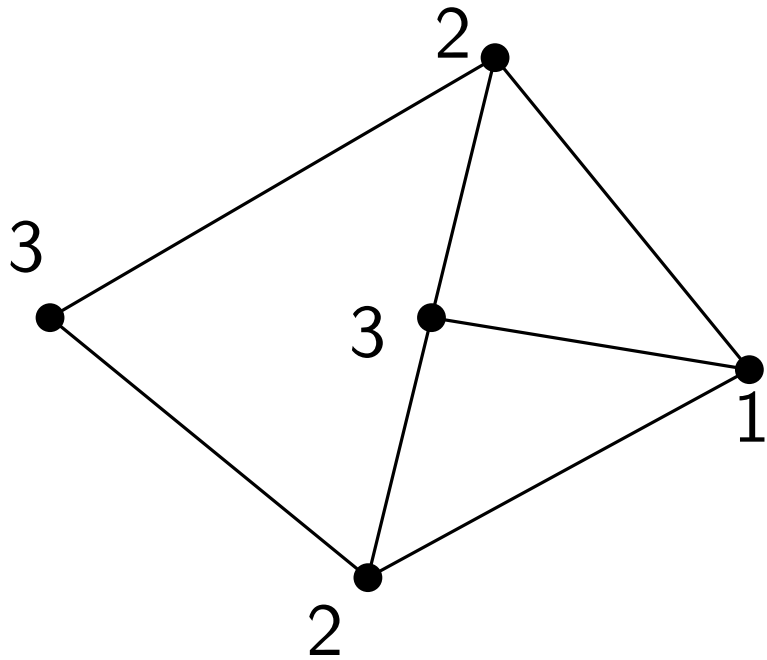
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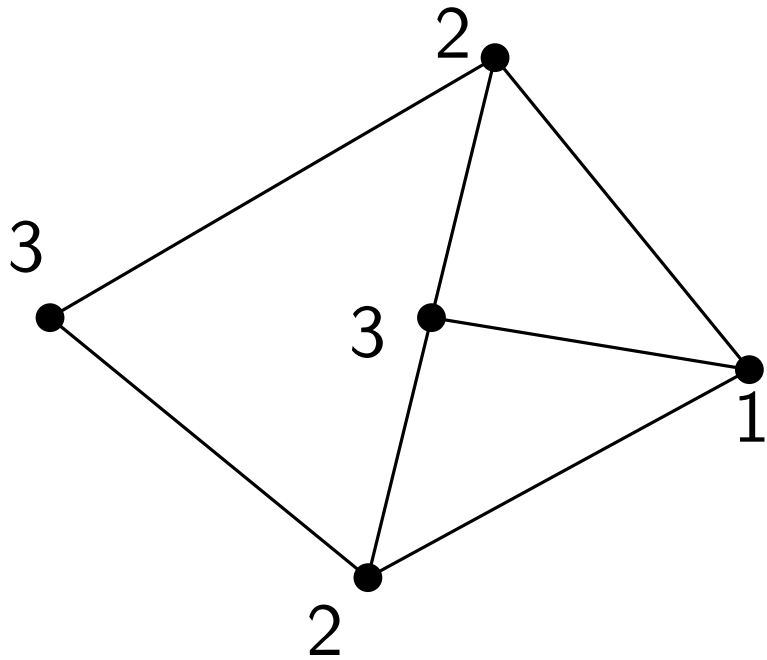
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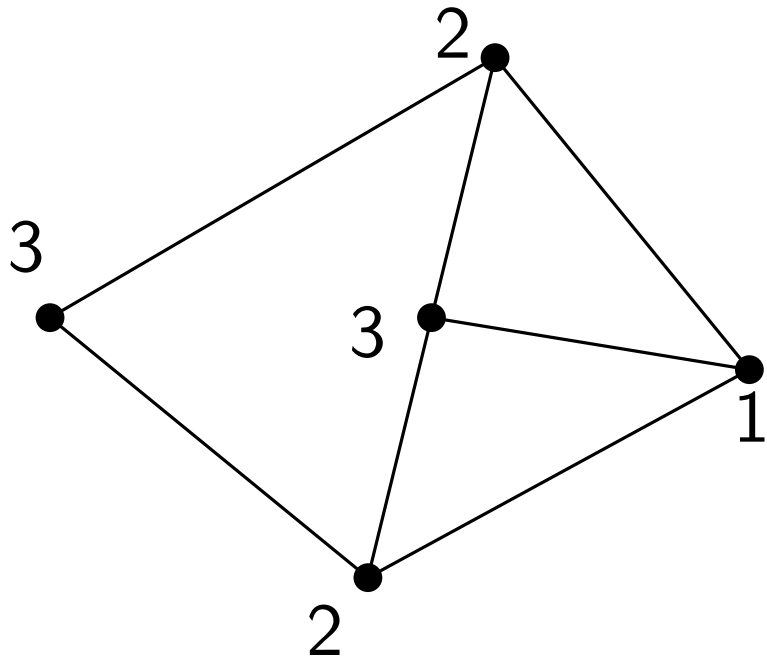


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- We will denote χ the *chromatic number* of Γ .
- $\chi_{\Gamma}(k) :=$ number of (proper) colorings of Γ with k colors

The algebra of symmetric functions

The algebra of symmetric functions

Given a set of variables $x = (x_1, x_2, \dots)$ we say a formal power series $f \in \mathbb{Z}[[x_1, x_2, \dots]]$ is a symmetric function if it satisfies:

- f has bounded degree
- $f(x_1, x_2, \dots) = f(x_{\sigma(1)}, x_{\sigma(2)}, \dots) \quad \forall \sigma \in \mathfrak{S}$

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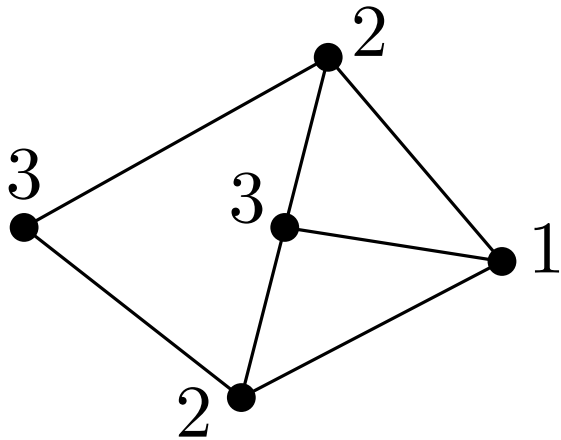
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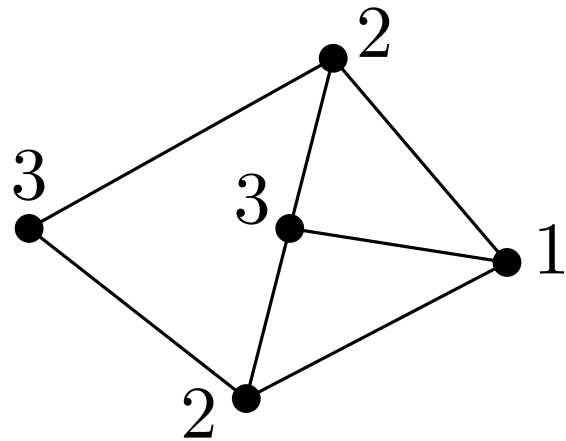
There are many different basis for Sym^n and it is an interesting combinatorial problem to find the transition matrix between them

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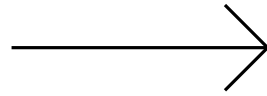
The chromatic symmetric function



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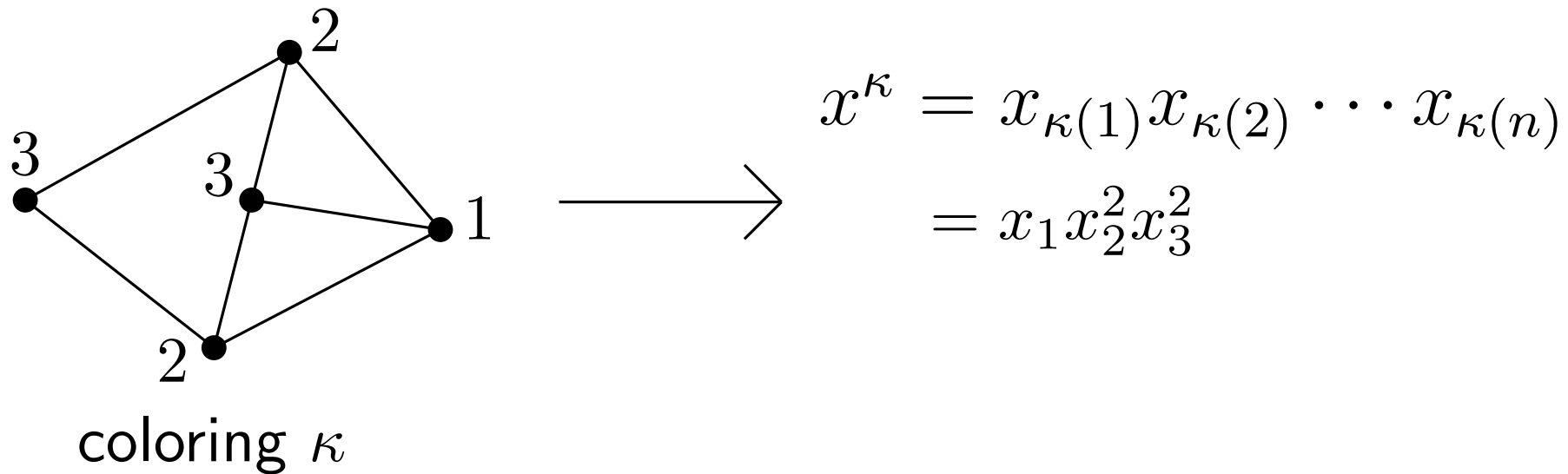


coloring κ



$$\begin{aligned} x^\kappa &= x_{\kappa(1)} x_{\kappa(2)} \cdots x_{\kappa(n)} \\ &= x_1 x_2^2 x_3^2 \end{aligned}$$

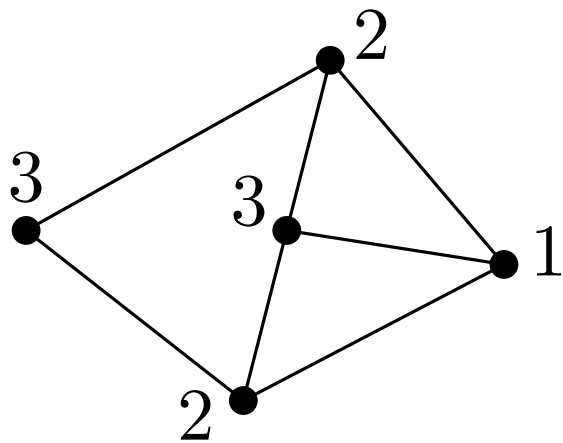
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In 1995 Richard P. Stanley defined the *chromatic symmetric function* of a graph Γ as:

$$X_\Gamma(x_1, x_2, \dots) = \sum_{\kappa \text{ proper}} x^\kappa \in \text{Sym}^{|V|}$$

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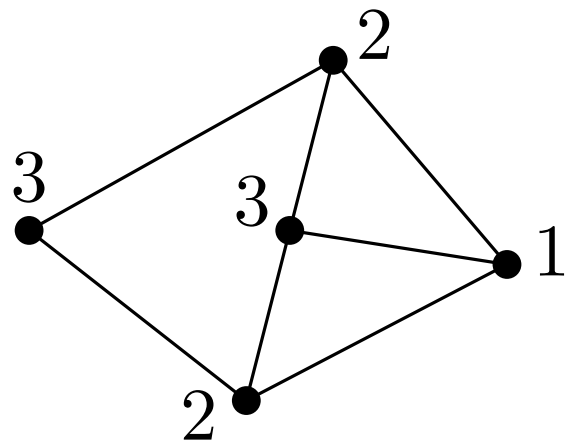
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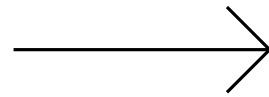
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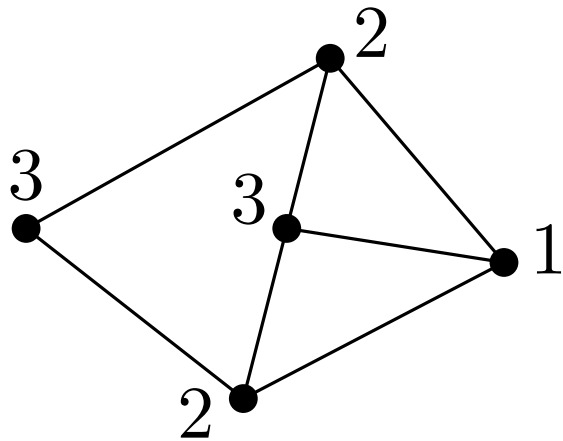
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Does X_Γ distinguish non isomorphic graphs?

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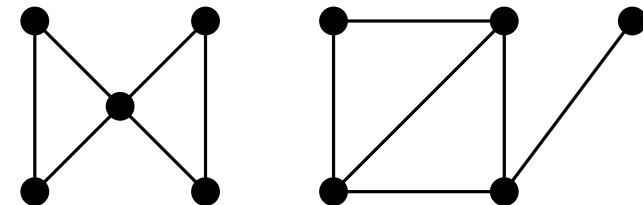
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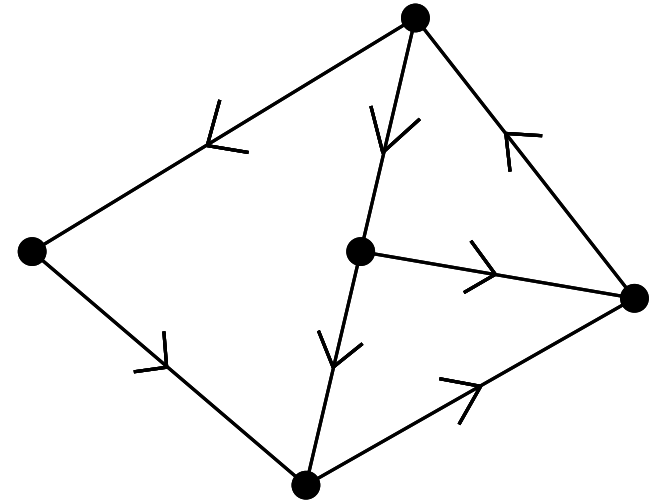
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Ellzey's quasisymmetric invariant

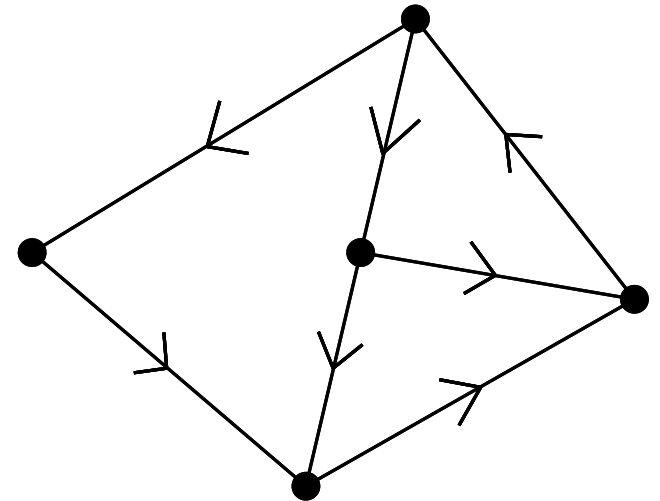
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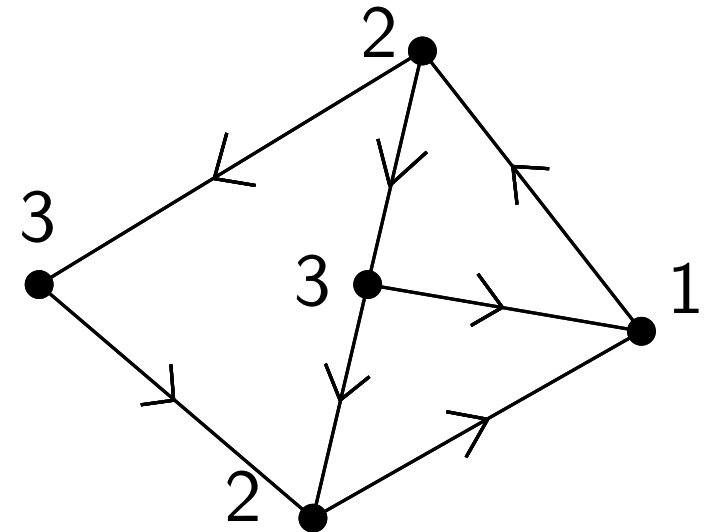
Given a directed graph $\vec{\Gamma}$ and a coloring κ of the underlying graph Γ we define an *ascent* of κ to be a directed edge $(i, j) \in E$ verifying $\kappa(i) < \kappa(j)$



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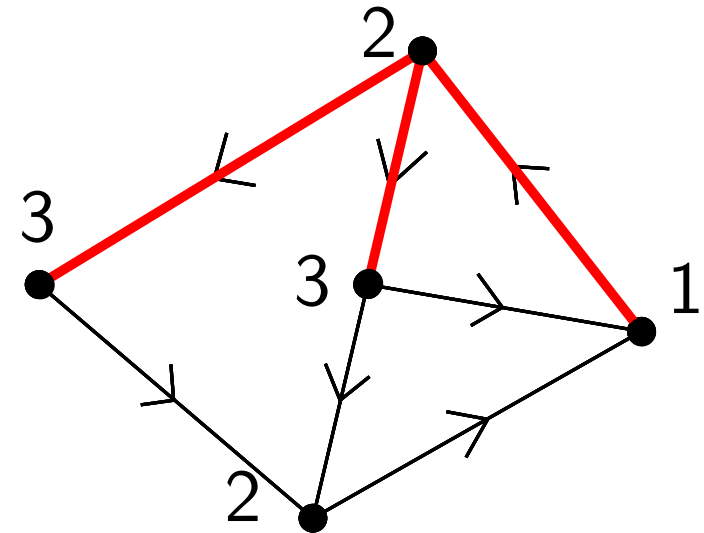
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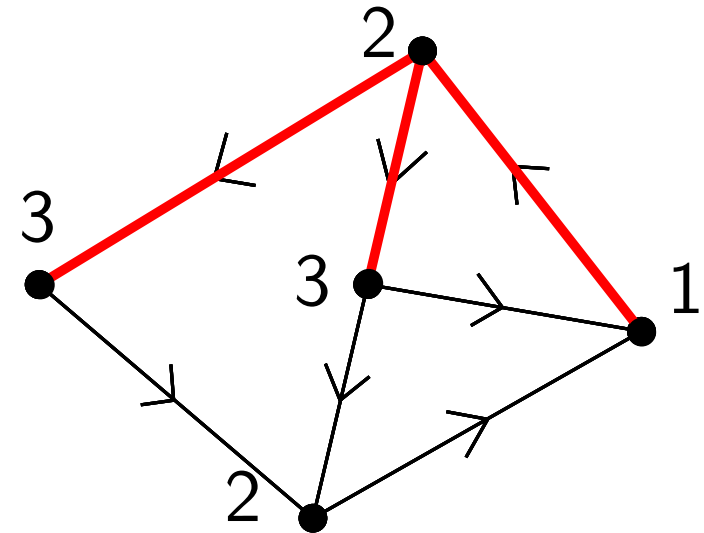


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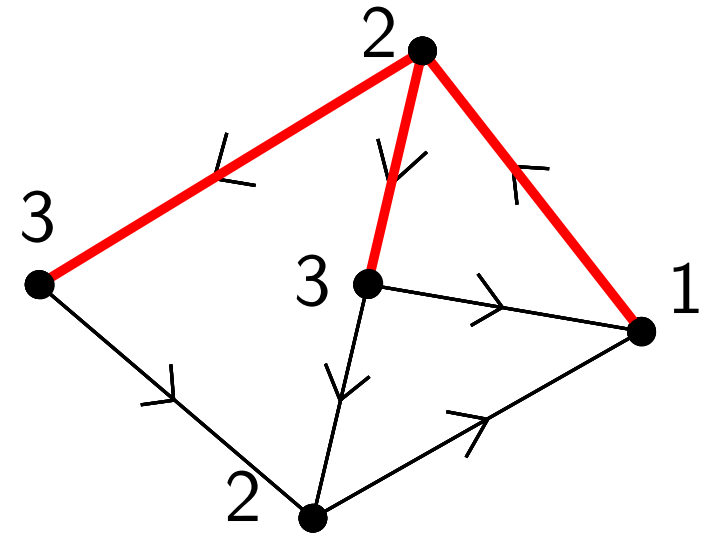
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Then the chromatic quasisymmetric function of a digraph is:

$$X_{\Gamma}(x; t) = \sum_{\kappa} x^{\kappa} \cdot t^{asc(\kappa)}$$

Quasisymmetric funtions

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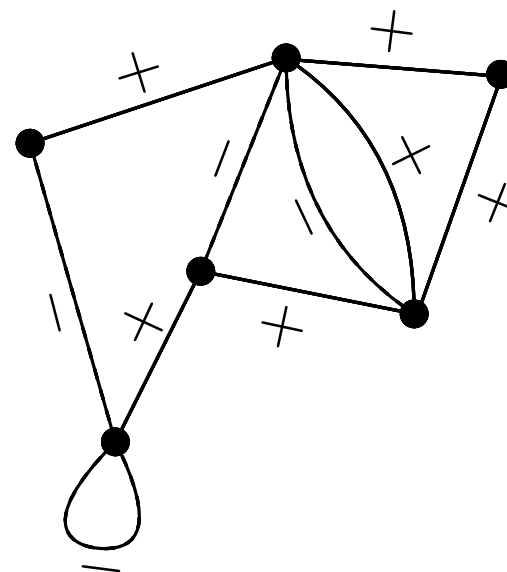
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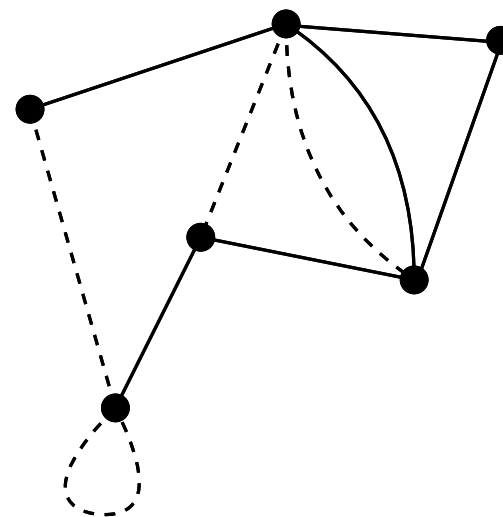
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- The bases of the quasisymmetric function algebra are indexed by compositions ($M_{\alpha}, F_{\alpha} \dots$)
- Quasisymmetric functions are a very important object on algebraic combinatorics

Signed graphs

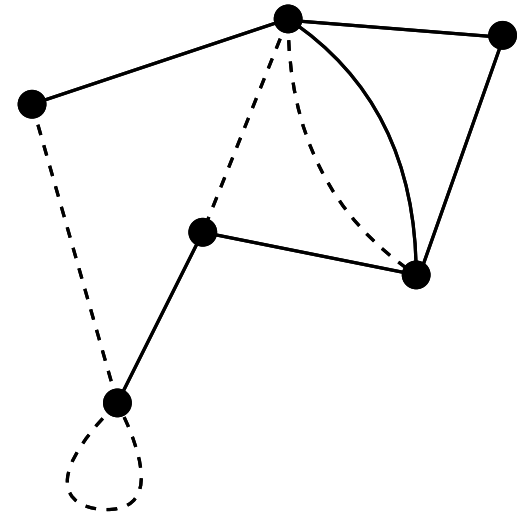


Signed graphs



Signed graphs

How do we color signed graphs?

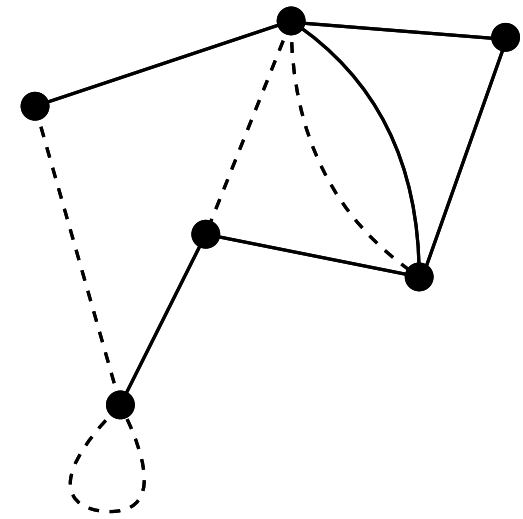


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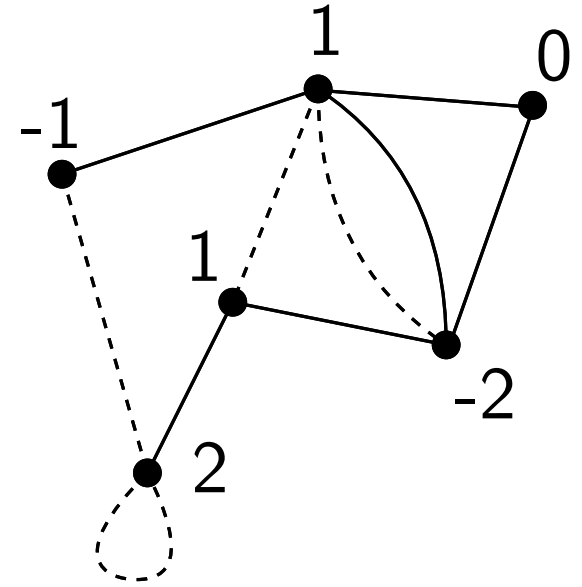


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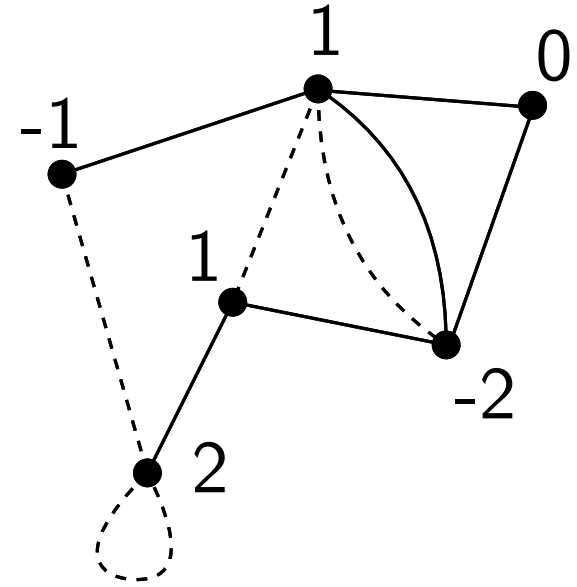


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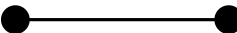
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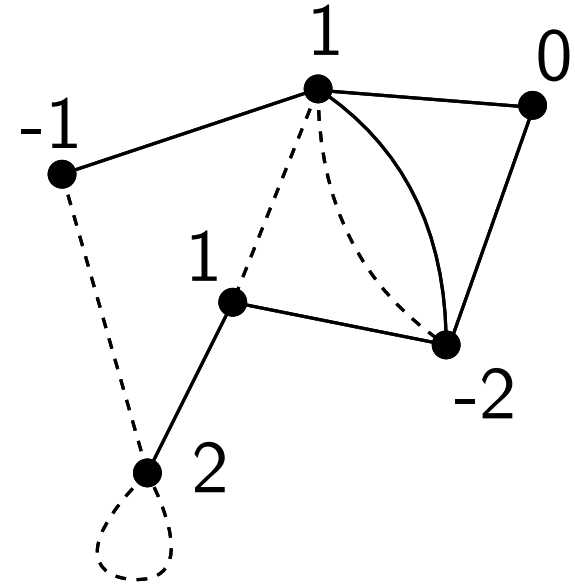
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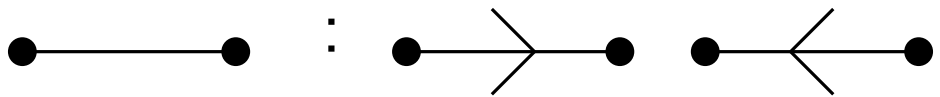
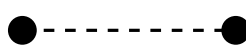
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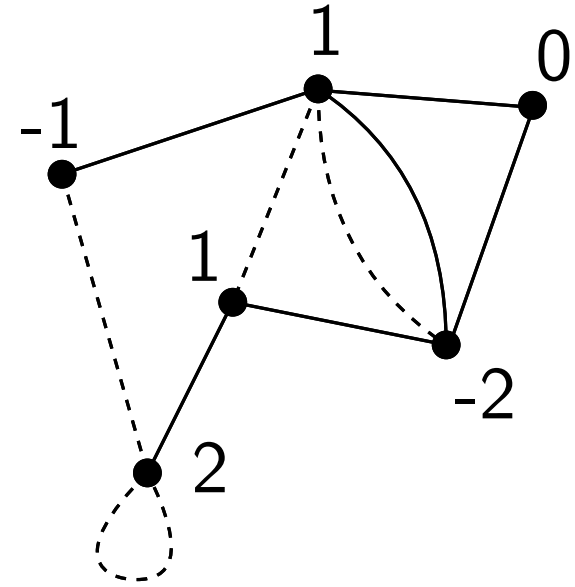
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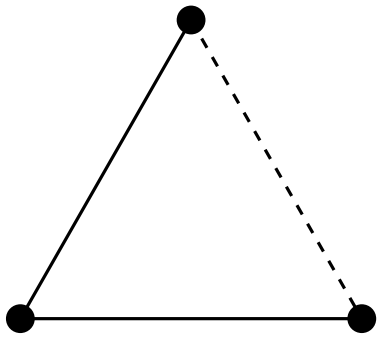
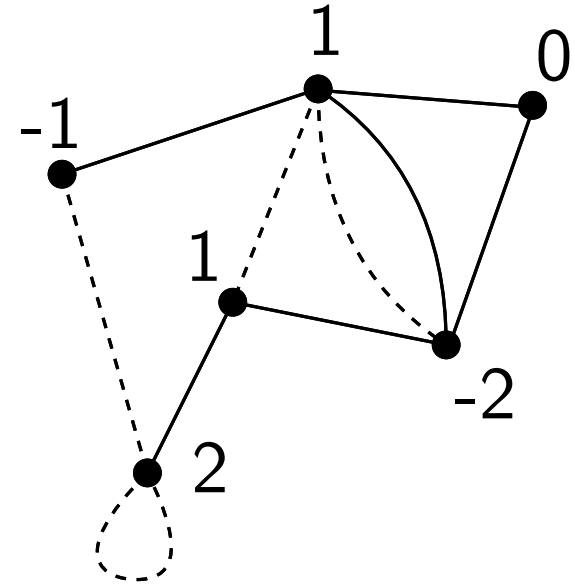
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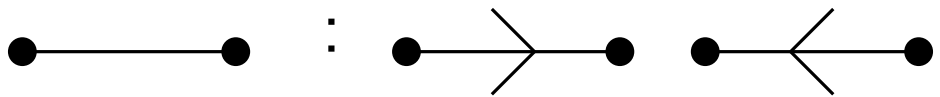
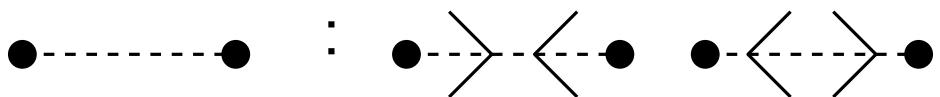
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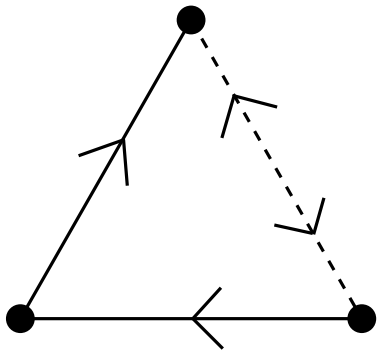
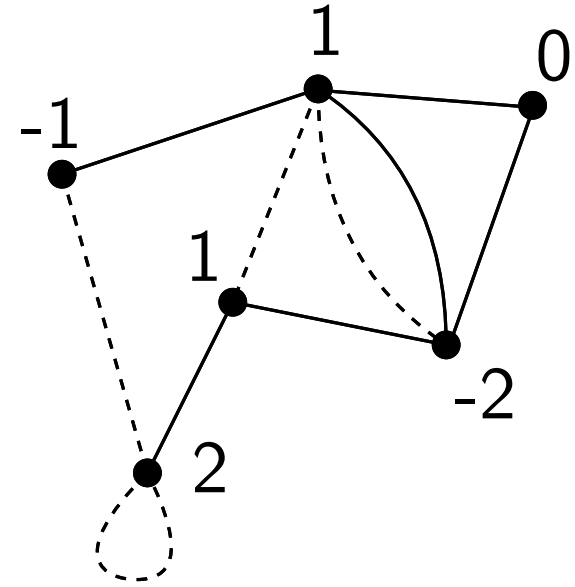
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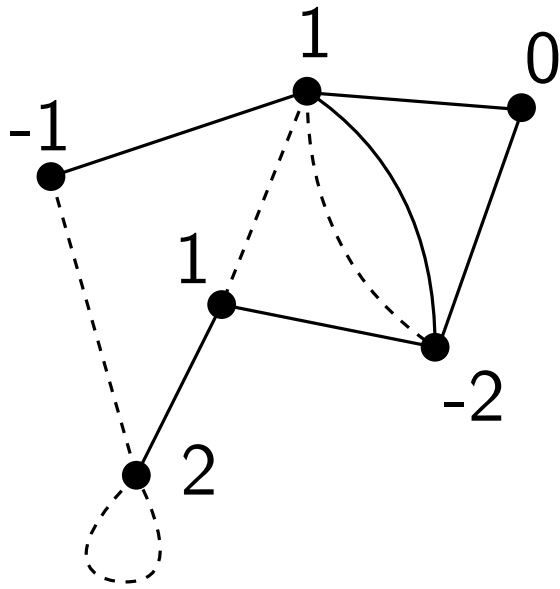


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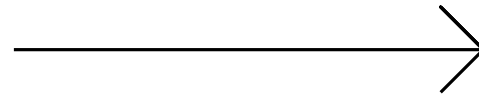
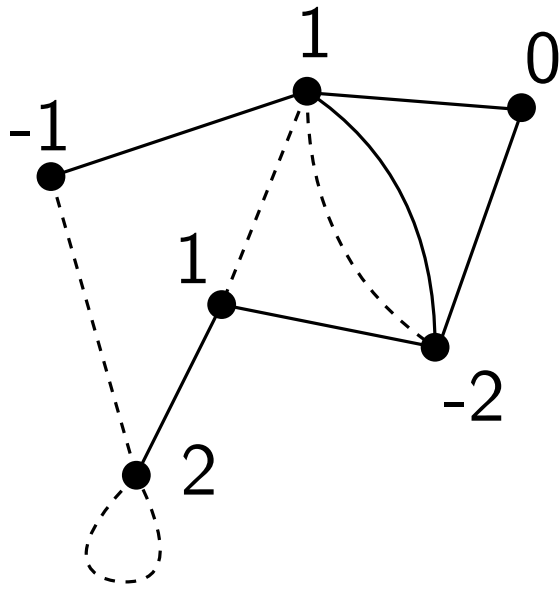
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Theory developed by Thomas Zaslavsky

The signed-symmetric invariant



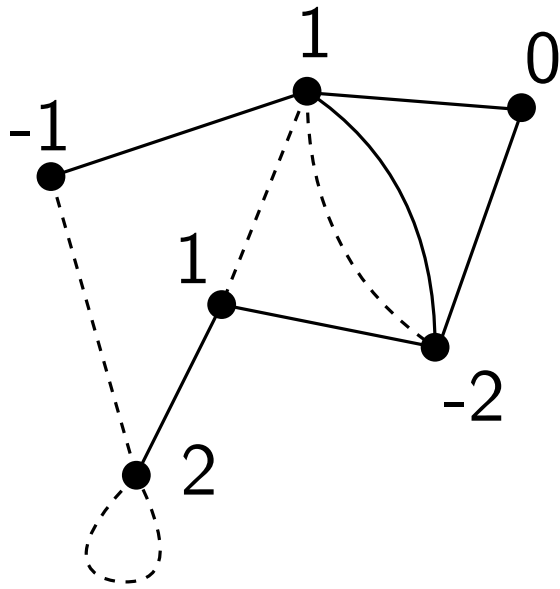
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Monomial in the set of
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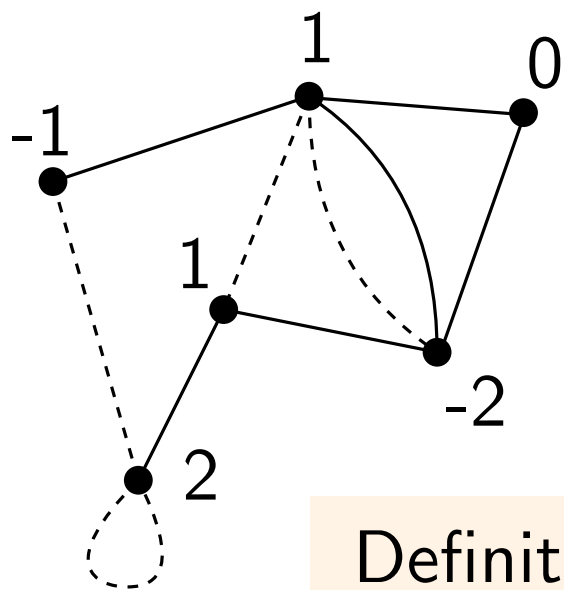


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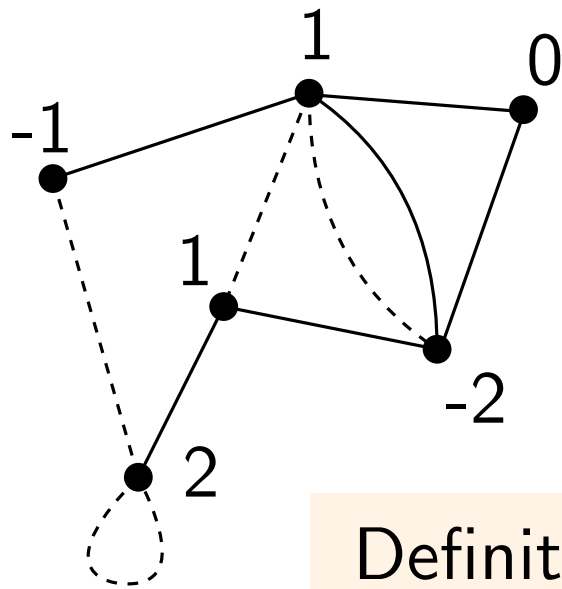
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Definition (Kuroda & Tsujie 2021)

Given a signed graph Σ we define the *signed-symmetric chromatic function* to be:

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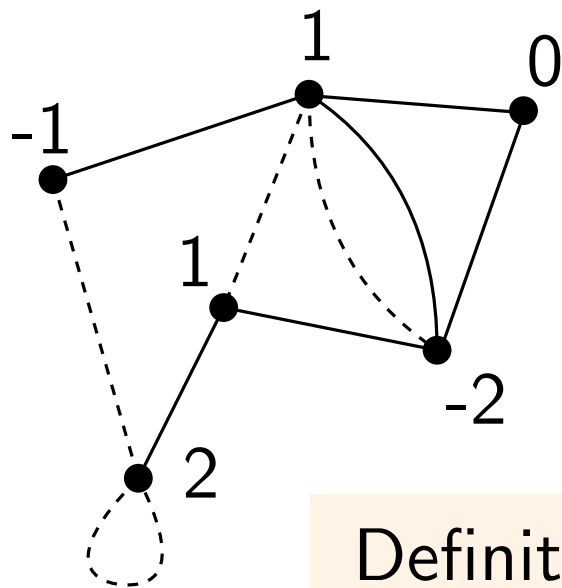
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Example



The signed-symmetric invariant



Monomial in the set of variables

$$\mathbf{x} = (\dots, x_{-1}, x_0, x_1, \dots)$$

$$\mathbf{x}^\kappa = x_0 x_1^2 x_{-1} x_2 x_{-2}$$

Definition (Kuroda & Tsujie 2021)

Given a signed graph Σ we define the *signed-symmetric chromatic function* to be:

$$X_\Sigma(\dots x_{-1}, x_0, x_1, \dots) = \sum_{\kappa \text{ proper}} \mathbf{x}^\kappa$$

Example



$$X_\Sigma = 2 \sum_{i,j \in \mathbb{N}, i < j} x_i x_j + 2 \sum_{i,j \in \mathbb{N}, i \neq j} x_i x_{-j}$$

$$2 \sum_{i,j \in \mathbb{N}, i < j} x_{-i} x_{-j} + x_0 \sum_{i \in \mathbb{N}} x_i + x_0 \sum_{i \in \mathbb{N}} x_{-i}$$

Signed-symmetric functions

$$S\mathfrak{S} := \{\text{bijections } \sigma : \mathbb{Z} \rightarrow \mathbb{Z} : \sigma(i) = -\sigma(-i) \ \forall i \in \mathbb{Z}\}$$

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We say a formal power series $f \in \mathbb{Z}[[\dots, x_{-1}, x_0, x_1, \dots]]$ is a *signed symmetric function* if:

- f has bounded degree
- $f(\dots, x_{-1}, x_0, x_1, \dots) = f(\dots, x_{\sigma(-1)}, x_{\sigma(0)}, x_{\sigma(1)} \dots) \ \forall \sigma \in S\mathfrak{S}$

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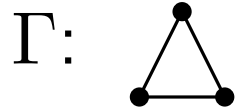
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$$X_{\Sigma}(\dots, x_{-1}, x_0, x_1, \dots) \in SSym^{|V|}$$

What are we working on?

Recap:

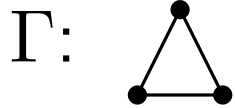
simple graph



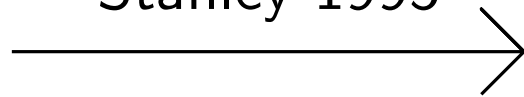
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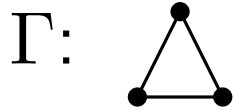


$X_{\Gamma}(x_1, x_2, \dots) \in Sym$

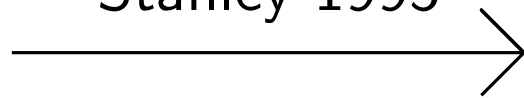
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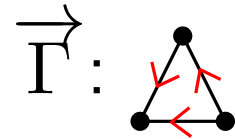


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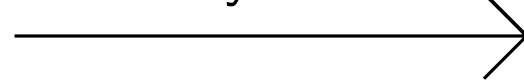


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directed graph



Ellzey 2017

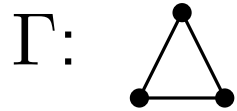


$$X_{\vec{\Gamma}}(x_1, x_2, \dots; \textcolor{red}{t}) \in QSym[t]$$

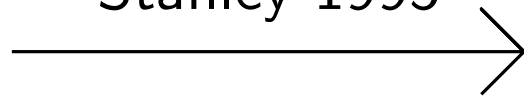
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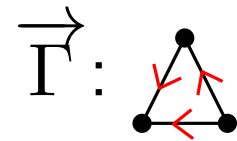


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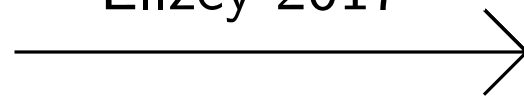


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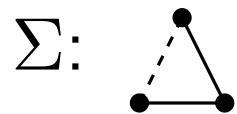


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$$X_{\vec{\Gamma}}(x_1, x_2, \dots; t) \in QSym[t]$$

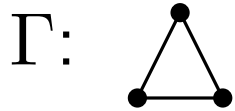
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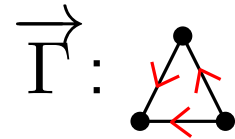
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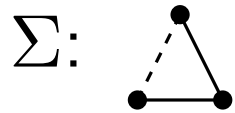
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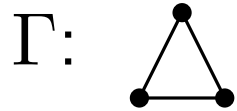
Kuroda 2021

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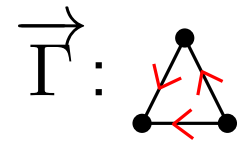
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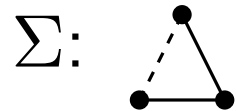
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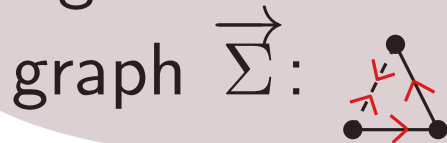
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This invariant belongs to $SQSym[t]$

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- We have given necessary conditions for $X_{\vec{\Sigma}}$ to be in $SSym[t]$

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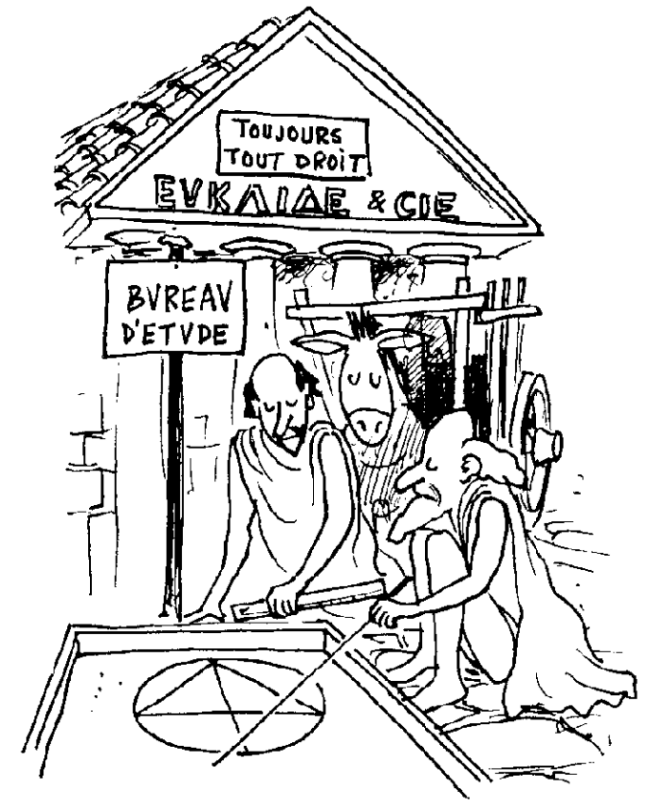
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The bibliography

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- **Kuroda & Tsujie** (2021). *Chromatic Signed-Symmetric Functions of Signed Graphs.* arXiv preprint
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- **Stanley** (1995). *A symmetric function generalization of the chromatic polynomial of a graph.* Adv. Math.
- **Zaslavsky** (1991). *Orientation of signed graphs.* Eur. J. Comb.

Thank you for your
attention!!

Eskerrik asko!



The geometric viewpoint

$$\left\{ \begin{array}{l} \text{Simple graphs} \\ \text{with } d \text{ vertices} \end{array} \right\} \longrightarrow \left\{ \begin{array}{l} \text{Hyperplane} \\ \text{arrangements} \\ \text{on } (\mathbb{R}^+)^d \end{array} \right\}$$

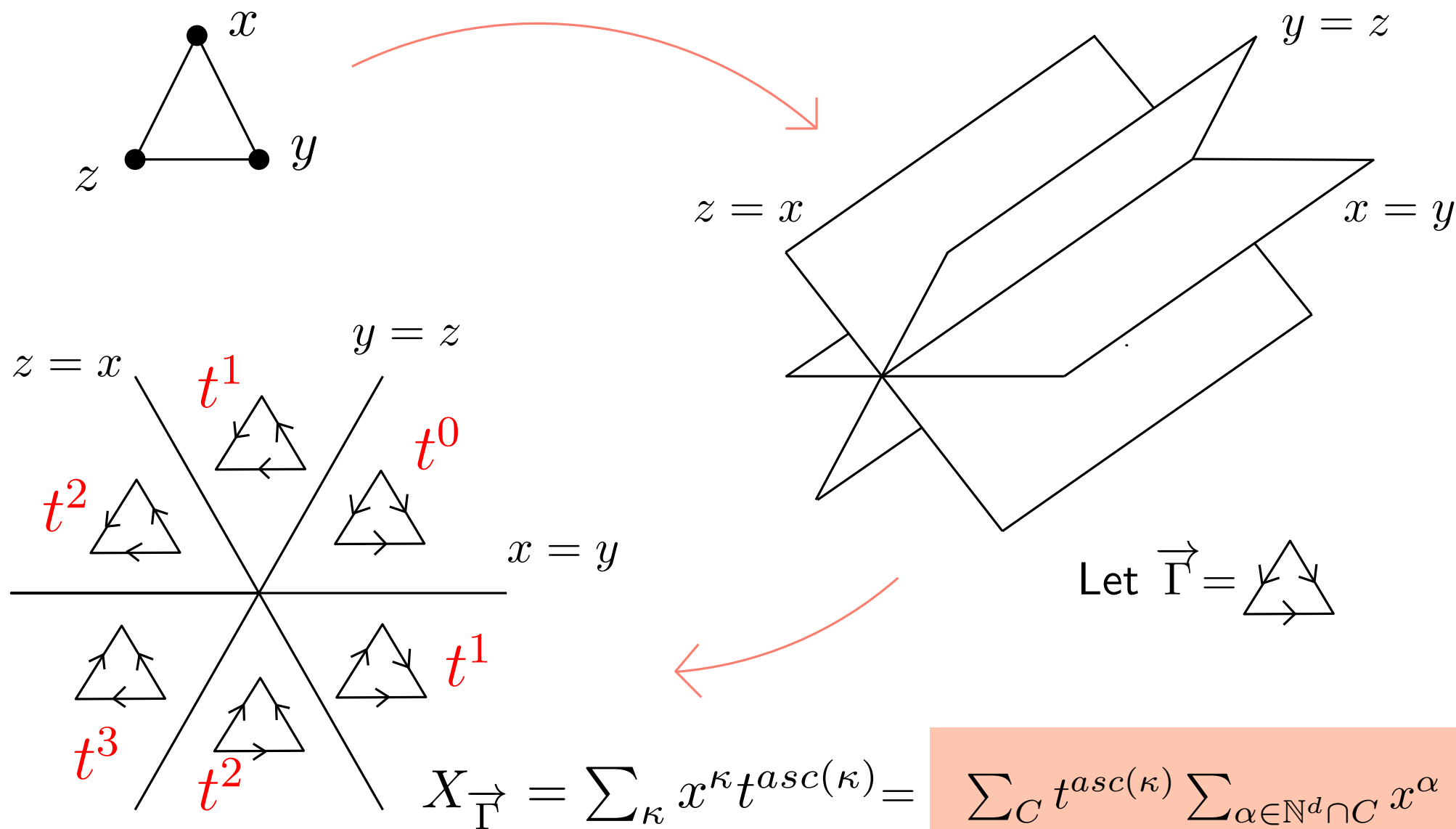
$$\Gamma = (V, E) \longmapsto \mathcal{H}_\Gamma = \{x_i = x_j : (i, j) \in E\}$$

$$\left\{ \begin{array}{l} \text{Proper colorings of } \Gamma \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{Integer points} \\ \text{in } (\mathbb{R}^+)^d \setminus \mathcal{H}_\Gamma \end{array} \right\}$$

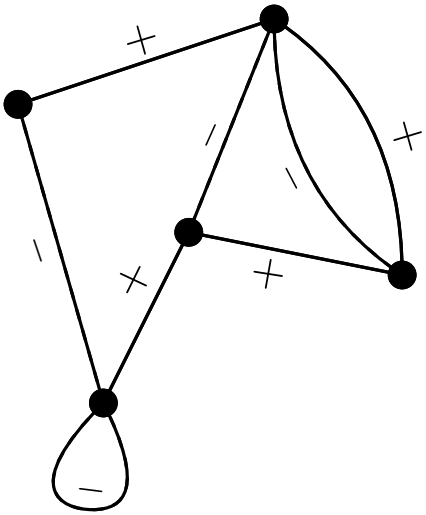
$$\kappa \longleftrightarrow (\kappa(1), \kappa(2), \dots, \kappa(d))$$

$$\left\{ \begin{array}{l} \text{Acyclic orientations} \\ \text{of } \Gamma \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{Connected} \\ \text{components of} \\ (\mathbb{R}^+)^d \setminus \mathcal{H}_\Gamma \end{array} \right\}$$

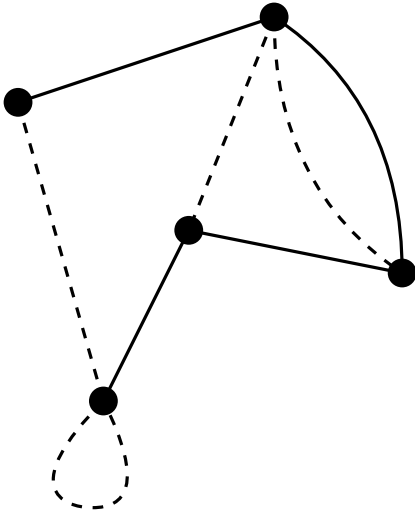
The geometric viewpoint: an example



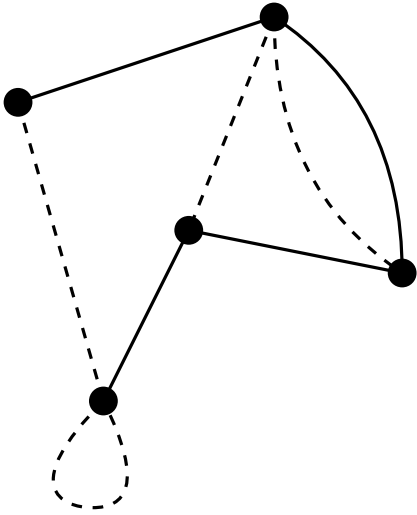
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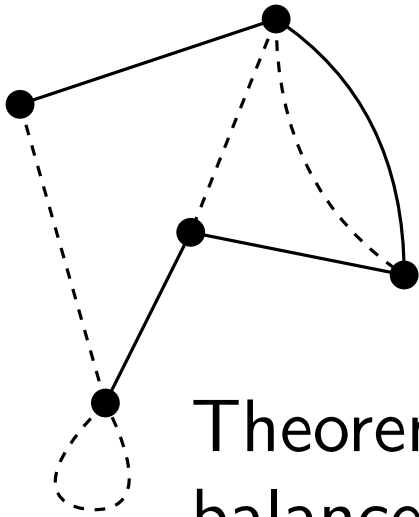


Signed graphs: definition



Definition: we say a signed graph Σ is *balanced* if every cycle has an even number of negative edges

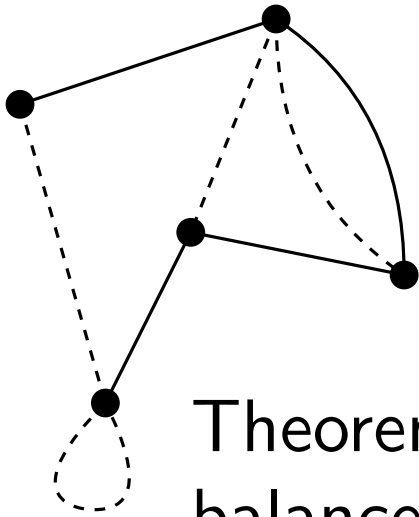
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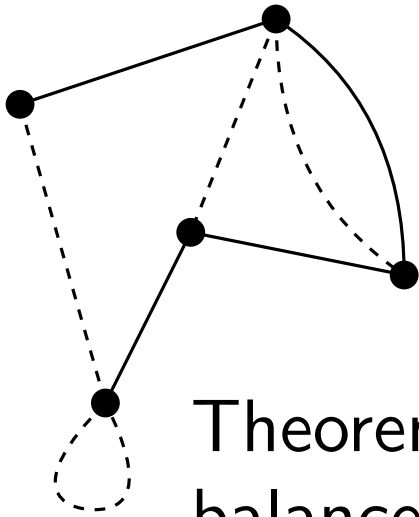


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- Every edge between points in A is positive
- Every edge between points in B is positive
- Every edge between points in A and B is negative

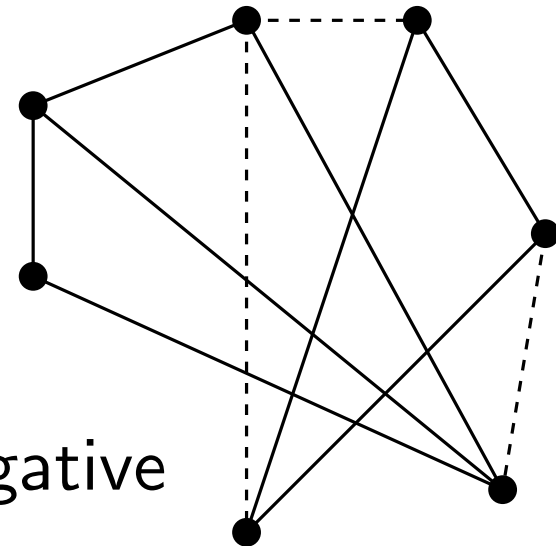
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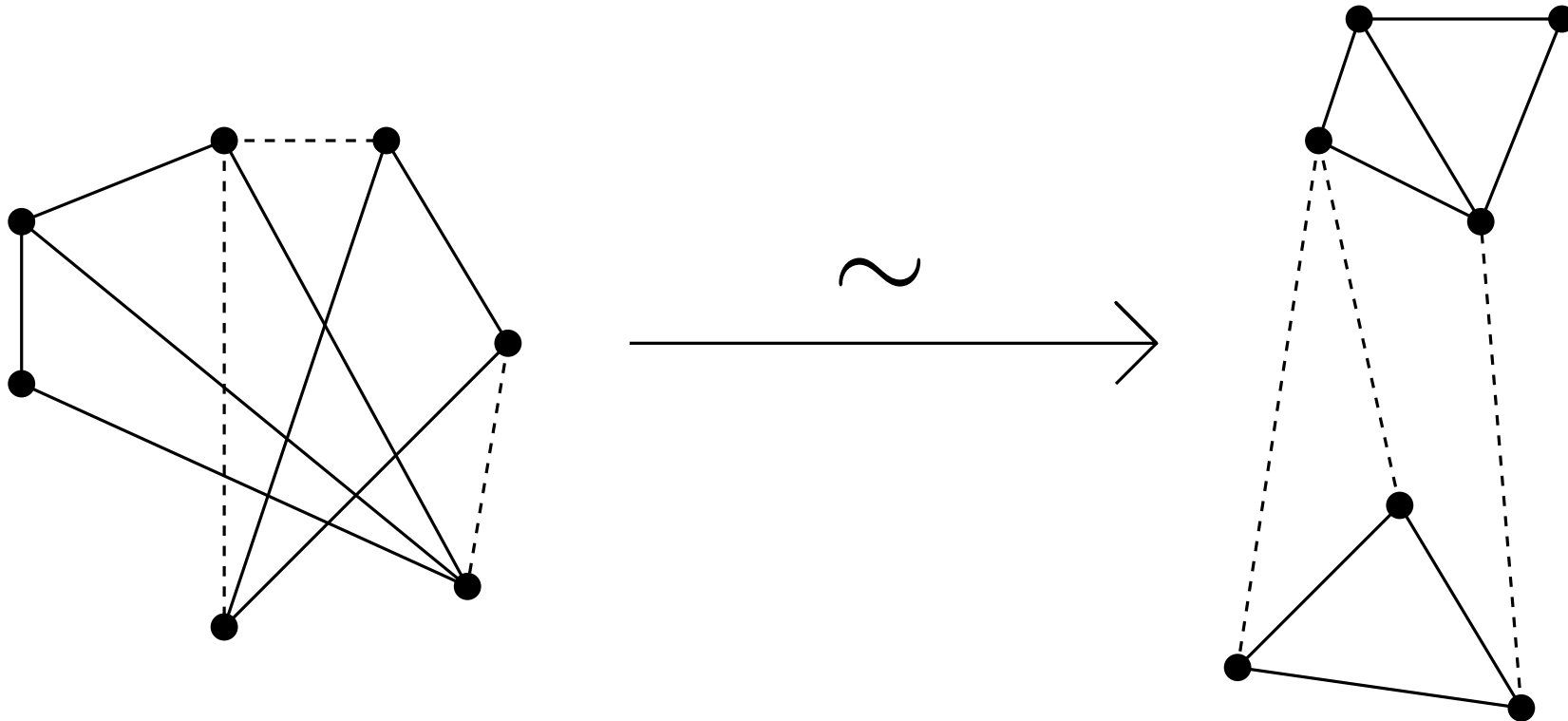
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Balanced graph characterization



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